

Practical 1: Protein Structure Manipulation in Python

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Repository: <https://github.com/VictorNoguera/Biophysics>

Exercise 1: Pairs of residues with CA atoms closer than a given distance

Execution & Input: `python "Practical 1.py" 1 1UBQ.pdb 8.0`

Output Excerpt:

```
None
MET 1 - GLN 2 : 3.80 A
MET 1 - ILE 3 : 6.58 A
MET 1 - GLU 16 : 5.51 A
MET 1 - VAL 17 : 5.36 A
(...)
LEU 73 - ARG 74 : 3.79 A
LEU 73 - GLY 75 : 6.02 A
LEU 73 - GLY 76 : 7.79 A
ARG 74 - GLY 75 : 3.80 A
ARG 74 - GLY 76 : 7.07 A
GLY 75 - GLY 76 : 3.79 A
```

Results & Conclusion:

The script calculates the distance between the main atoms (Carbon Alpha) of all amino acids. It shows which pairs are closer than 8.0 Å, helping us see which parts of the protein are physically near each other in 3D space.

Exercise 2: List of all atoms for a given residue number

Execution & Input: `python "Practical 1.py" 2 1UBQ.pdb A 24`

Output:

```
None
Atom: N      | Coordinates: X=33.548, Y=21.526, Z=16.950
Atom: CA     | Coordinates: X=35.031, Y=21.722, Z=17.069
Atom: C      | Coordinates: X=35.615, Y=22.190, Z=15.759
Atom: O      | Coordinates: X=36.532, Y=23.046, Z=15.724
```

```
Atom: CB    | Coordinates: X=35.667, Y=20.383, Z=17.447
Atom: CG    | Coordinates: X=37.128, Y=20.293, Z=17.872
Atom: CD    | Coordinates: X=37.561, Y=18.851, Z=18.082
Atom: OE1   | Coordinates: X=37.758, Y=18.024, Z=17.195
Atom: OE2   | Coordinates: X=37.628, Y=18.599, Z=19.313
```

Results & Conclusion:

The code searches for a specific amino acid (Glutamic Acid at position 24 in Chain A). It simply finds it and lists the exact X, Y, and Z coordinates for all 9 of its atoms.

Exercise 3: Determine all possible hydrogen bonds

Execution & Input: `python "Practical 1.py" 3 1UBQ.pdb 3.5`

Output Excerpt:

```
None
MET1 (N) -- VAL17 (O) : 2.56 A
MET1 (N) -- HOH98 (O) : 3.02 A
MET1 (N) -- HOH110 (O) : 2.37 A
MET1 (O) -- GLN2 (N) : 2.26 A
(...)
HOH101 (O) -- HOH106 (O) : 3.29 A
HOH113 (O) -- HOH130 (O) : 2.60 A
HOH114 (O) -- HOH122 (O) : 2.15 A
HOH117 (O) -- HOH124 (O) : 3.49 A
HOH121 (O) -- HOH128 (O) : 2.11 A
```

Results & Conclusion:

The program filters out everything except polar atoms (Oxygen, Nitrogen, Sulfur). It finds pairs that are less than 3.5 Å apart, which strongly suggests they are forming hydrogen bonds to keep the protein stable.

Exercise 4: CA atoms of a given residue type

Execution & Input: `python "Practical 1.py" 4 1UBQ.pdb ALA`

Output:

None

Chain A, Residue 28 | Coordinates: X=38.79, Y=25.76, Z=13.88

Chain A, Residue 46 | Coordinates: X=18.44, Y=29.14, Z=19.08

Results & Conclusion:

The script searches the entire protein looking only for Alanines (ALA). It correctly finds that there are exactly two (at positions 28 and 46) and prints the 3D coordinates of their central atoms.